We can represent the Kalman filter in the form

$$m_{t+1} = m_t + K_{t+1} a_{t+1}$$

which implies that

$$E\left(m_{t+1} - m_t\right)^2 = K_{t+1}^2 \sigma_{a,t+1}^2$$

where $a_{t+1} = y_{t+1} - m_t$ and the variance of $a_t$ is equal to $\sigma_{a,t+1}^2 = (\Sigma_{t+1} + R)$ from equation (5.7.5). This implies

$$E\left(m_{t+1} - m_t\right)^2 = \frac{\Sigma_{t+1}^2}{\Sigma_{t+1} + R}.$$

For the purposes of our discrete-time counterpart of the Jovanovic model in chapter 7, it will be convenient to represent the motion of $m_{t+1}$ by means of the equation

$$m_{t+1} = m_t + g_{t+1} u_{t+1}$$

where $g_{t+1} \equiv \left(\frac{\Sigma_{t+1}^2}{\Sigma_{t+1}+R}\right)^{.5}$ and $u_{t+1}$ is i.i.d. normalized and standardized with mean zero and variance 1 constructed to obey $g_{t+1} u_{t+1} \equiv K_{t+1} a_{t+1}$.

## 6.4. LQ permanent income model

This section describes a linear quadratic savings problem whose solution is a rational expectations version of the permanent income model of Friedman (1957) and Hall (1978). We use this model as a vehicle for illustrating impulse response functions, alternative choices of the state, the idea of cointegration, and an invariant subspace method.

The LQ permanent income model is a modification (and not quite a special case, for reasons that will be apparent later) of the following incomplete markets "savings problem" to be studied in chapter 18. A consumer has preferences over consumption streams that are ordered by the utility functional

$$E_0 \sum_{t=0}^{\infty} \beta^t u\left(c_t\right) \tag{6.4.1}$$

where $E_t$ is the mathematical expectation conditioned on the consumer's time $t$ information, $c_t$ is time $t$ consumption, $u(c)$ is a strictly concave one-period

utility function, and $\beta \in (0,1)$ is a discount factor. The consumer maximizes (6.4.1) by choosing a consumption, borrowing plan $\{c_t, b_{t+1}\}_{t=0}^{\infty}$ subject to the sequence of budget constraints

$$c_t + b_t = R^{-1} b_{t+1} + y_t \tag{6.4.2}$$

where $\{y_t\}$ is an exogenous stationary endowment process, $R$ is a constant gross risk-free interest rate, $b_t$ is one-period risk-free bond maturing at $t$, and $b_0$ is a given initial condition. It is important to note that for $t \geq 1$, $b_t$ is chosen at time $t-1$ and that $b_0$ is a given initial condition at time $t = 0$. We shall assume that $R^{-1} = \beta$. We assume that the endowment process has the state-space representation

$$z_{t+1} = \check{A} z_t + \check{C} w_{t+1} \tag{6.4.3a}$$

$$y_t = \check{G} z_t \tag{6.4.3b}$$

where $w_{t+1}$ is an i.i.d. process with mean zero and identity contemporaneous covariance matrix, $\check{A}$ is a stable matrix, its eigenvalues being strictly below unity in modulus, and $\check{G}$ is a vector. The *state* vector confronting the household at $t$ is $[\, b_t \quad z_t \,]'$, where $b_t$ is its one-period debt falling due at the beginning of period $t$ and $z_t$ contains all variables useful for forecasting its future endowment. The consumer's choices $c_t$ and $b_{t+1}$ at $t$ are constrained by budget constraint (6.4.2) and by a limit on how fast and far borrowing $R^{-1} b_{t+1}$ is permitted to rise. Without limits on borrowing, the consumer could run a Ponzi scheme that would allow him to issue enough debt to consume as much as he wants each period. A consumption plan that satisfies the budget constraint (6.4.2) and the limits on borrowing satisfies the first-order conditions[4]

$$E_t u'(c_{t+1}) = u'(c_t), \quad \forall t \geq 0 \tag{6.4.4}$$

at interior solutions for consumption.

To make the problem fit into the setting of our linear-quadratic permanent income structure, we now assume the quadratic utility function $u(c_t) = -.5(c_t -$

[4] We shall study how to derive this first-order condition in detail as well as alternative specifications of borrowing limits in later chapters.

$\gamma)^2$, where $\gamma > 0$ is a bliss level of consumption and we allow $c_t$ to be negative.[5] For our linear quadratic savings problem, we impose the following condition on the consumption, borrowing plan:

$$E_0 \sum_{t=0}^{\infty} \beta^t b_t^2 < +\infty. \tag{6.4.5}$$

This condition suffices to rule out Ponzi schemes and serves as the limits on borrowing in our model. Furthermore, the quadratic utility function and allowing consumption to be negative imply that Euler equation (6.4.4) becomes[6]

$$E_t c_{t+1} = c_t. \tag{6.4.6}$$

To deduce the optimal decision rule, we want to solve the system of difference equations formed by (6.4.2) and (6.4.6) subject to the boundary condition (6.4.5). To assist us will exploiting the fact that the solution to this linear quadratic optimization problem satisfies a *certainty equivalence* property described in section **@sec:certequiv@** of chapter 5. That principle allows us to solve the consumer's problem in the following way.[7] Temporarily pretend that non-financial income $\{y_t\}_{t=0}^{\infty}$ is known for sure and compute an optimal

Iterating on the grammar here: replaced 'exploit' by 'exploiting' (You'll be the final judge :-)

[5] We allow the consumer actually to produce rather than to consume by setting a negative value of $c_t$. For a given $c < 0$, the produced amount of goods is $\tilde{y} = -c > 0$ and the marginal disutility of producing additional goods is $\partial u(-\tilde{y})/\partial \tilde{y} = -(\tilde{y} + \gamma) < 0$. The consumer can use the produced good to pay off any debt falling due that period or to lend to others and receive their repayments next period. For another example of interconnected consumption and production activities, see Kiyotaki och Wright's (1993) search model of money and differentiated goods in chapter 30. Kiyotaki and Wright assume that each consumer can produce a good that other consumers want but the the producer himself or herself does not. That induces each producer to want to exchange his good for one produced by someone else, either directly in a bilateral exchange or indirectly by accumulating a good that function as a medium of exchange called money.

[6] A linear marginal utility is essential for deriving (6.4.6) from (6.4.4). Suppose instead that we had imposed the following more standard assumptions on the utility function: $u'(c) > 0, u''(c) < 0, u'''(c) > 0$ and required that $c \geq 0$. The Euler equation remains (6.4.4). But $u''' > 0$ implies via Jensen's inequality that $E_t u'(c_{t+1}) > u'(E_t c_{t+1})$. This inequality together with (6.4.4) implies that $E_t c_{t+1} > c_t$ (consumption is said to be a 'submartingale'), so that consumption stochastically diverges to $+\infty$. The consumer's savings also diverge to $+\infty$. Chapter 18 discusses such divergence in a class of 'precautionary savings' models.

[7] Section I of the introduction to Lucas and Sargent (1981) describes how the certainty equivalence property applies in a class of linear-quadratic problems that includes ours here. See especially pages xiv-xv.

consumption plan under that perfect-foresight assumption that expresses time $t$ consumption $c_t$ as a function of $b_t$ and $\{y_{t+j}\}_{j=0}^{\infty}$. The certainty equivalence property allows us then to compute an optimal plan under uncertainty about the non-financial income process by simply replacing $\{y_{t+j}\}_{j=0}^{\infty}$ with $\{E_t y_{t+j}\}_{j=0}^{\infty}$, where $E_t(\cdot)$ denotes a mathematical expectation conditional on the consumer's time $t$ information about the tail $\{y_{t+j}\}_{j=0}^{\infty}$ of the non-financial income process.

To implement this certainty equivalence logic, at time $t$ consider one possible continuation path $\{y_{t+j}\}_{j=0}^{\infty}$ of non-financial income and the implied optimal choices of $\{c_{t+j}, b_{t+1+j}\}_{j=0}^{\infty}$. For that particular future path, solve (6.4.2) forward and impose $\lim_{T\to+\infty} \beta^T b_{T+1} = 0$ to get

$$b_t = \sum_{j=0}^{\infty} \beta^j \left(y_{t+j} - c_{t+j}\right). \tag{6.4.7}$$

Imposing $\lim_{T\to+\infty} \beta^T b_{T+1} = 0$ suffices to impose (6.4.5) on the debt path. Consider repeating this calculation for *all* possible realizations of continuation paths $\{y_{t+j}\}_{j=0}^{\infty}$ of non-financial income that appear on the right side of (6.4.7). Take a weighted sum over all those paths, using conditional probabilities as weights to deduce

$$b_t = \sum_{j=0}^{\infty} \beta^j \left(E_t y_{t+j} - E_t c_{t+j}\right). \tag{6.4.8}$$

Since conditional probabilities sum to one, the left sides of expressions (6.4.7) and (6.4.8) are equal.[8] Next, use (6.4.6) and the law of iterated expectations to deduce

$$b_t = \sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \frac{1}{1-\beta} c_t \tag{6.4.9}$$

or

$$c_t = (1-\beta)\left[\sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - b_t\right]. \tag{6.4.10}$$

If we define the net rate of interest $r$ by $\beta = \frac{1}{1+r}$, we can also express this equation as

$$c_t = \frac{r}{1+r}\left[\sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - b_t\right]. \tag{6.4.11}$$

[8] Although equation (6.4.8) also holds for a more general optimal consumption, savings problem in which $u(c)$ is not quadratic, a solution strategy based on certainty-equivalence will not work for those problems.

Equation (6.4.10) or (6.4.11) asserts that consumption equals $\frac{r}{1+r}$ times the sum of nonfinancial wealth $\sum_{j=0}^{\infty} \beta^j E_t y_{t+j}$ and financial wealth $-b_t$. We can use a version of our formula (2.4.17) in (6.4.10) or (6.4.11) to obtain

$$c_t = (1-\beta)\left[\check{G}\left(I-\beta\check{A}\right)^{-1} z_t - b_t\right]. \tag{6.4.12}$$

This is a "consumption function" that represents $c_t$ as a function of the *state* $[b_t, z_t]$ that confronts the household, where from (6.4.3) $z_t$ contains present information that is useful for forecasting the endowment process.

Pulling together our preceding results, we can regard $(z_t, b_t)$ as the time $t$ state, where $z_t$ is an *exogenous* component of the state and $b_t$ is an *endogenous* component of the state vector. The system can be represented as

$$\begin{aligned} z_{t+1} &= \check{A} z_t + \check{C} w_{t+1} \\ b_{t+1} &= b_t + \check{G}\left[\left(I-\beta\check{A}\right)^{-1}\left(\check{A}-I\right)\right] z_t \\ y_t &= \check{G} z_t \\ c_t &= (1-\beta)\left[\check{G}\left(I-\beta\check{A}\right)^{-1} z_t - b_t\right]. \end{aligned}$$

where the last equation comes from applying to decision rule (6.4.11) the appropriate version our formula (2.4.17) for forecasting geometric sums of future values of the random variable $y_t$.

### *6.4.1. Another representation*

Another way to understand the system under the optimal decision rules for $(b_{t+1}, c_t)$ is to recognize that there is a point of view that allows us to regard the state as being $c_t$ together with $z_t$ and to regard $b_t$ as an outcome. Hall (1978) showed that this is a sharp way to summarize an LQ permanent income theory. We proceed to transform the state vector in this way.

To represent an optimal plan for $b_t$, substitute (6.4.10) into (6.4.2) and after rearranging obtain

$$b_{t+1} = b_t + \left(\beta^{-1} - 1\right) \sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \beta^{-1} y_t. \tag{6.4.13}$$

Next, shift (6.4.10) forward one period and eliminate $b_{t+1}$ by using (6.4.2) to obtain

$$c_{t+1} = (1-\beta)\sum_{j=0}^{\infty} E_{t+1}\beta^j y_{t+j+1} - (1-\beta)\left[\beta^{-1}(c_t + b_t - y_t)\right].$$

If we add and subtract $\beta^{-1}(1-\beta)\sum_{j=0}^{\infty}\beta^j E_t y_{t+j}$ from the right side of the preceding equation and rearrange, we obtain

$$c_{t+1} - c_t = (1-\beta)\sum_{j=0}^{\infty}\beta^j\left(E_{t+1}y_{t+j+1} - E_t y_{t+j+1}\right). \tag{6.4.14}$$

The right side is $(1-\beta)$ times the time $t+1$ innovation to the expected present value of the endowment process $\{y_t\}$. It is useful to express this innovation in terms of a moving average representation for income $y_t$. Suppose that the endowment process has the moving average representation[9]

$$y_{t+1} = d(L)\, w_{t+1} \tag{6.4.15}$$

where $\{w_{t+1}\}$ is an i.i.d. vector process with $Ew_{t+1} = 0$ and contemporaneous covariance matrix $Ew_{t+1}w'_{t+1} = I$, $d(L) = \sum_{j=0}^{\infty} d_j L^j$, where $L$ is the lag operator, and the household has an information set $w^t = [w_t, w_{t-1}, \ldots,]$ at time $t$.[10] Then notice that

$$y_{t+j} - E_t y_{t+j} = d_0 w_{t+j} + d_1 w_{t+j-1} + \cdots + d_{j-1} w_{t+1}.$$

It follows that

$$E_{t+1}y_{t+j} - E_t y_{t+j} = d_{j-1} w_{t+1}. \tag{6.4.16}$$

Using (6.4.16) in (6.4.14) gives

$$c_{t+1} - c_t = (1-\beta)\, d(\beta)\, w_{t+1}. \tag{6.4.17}$$

The object $d(\beta)$ is the present value of the moving average coefficients in the representation for the endowment process $y_t$. Furthermore $d(\beta)w_{t+1}$ is the time $t+1$ innovation to the expected present value of the endowment process $\{y_t\}$.

---

9 Representation (6.4.3) implies that $d(L) = \check{G}(I - \check{A}L)^{-1}\check{C}$.

10 A moving average representation for a process $y_t$ is said to be *fundamental* if the linear space spanned by $y^t$ is equal to the linear space spanned by $w^t$. A time-invariant innovations representation, attained via the Kalman filter as in section 2.7, is by construction fundamental.

After all of this work, we can represent the optimal decision rule for $c_t, b_{t+1}$ in the form of the two equations (6.4.14) and (6.4.9), which we repeat here for convenience:

$$c_{t+1} = c_t + (1-\beta)\sum_{j=0}^{\infty}\beta^j\left(E_{t+1}y_{t+j+1} - E_t y_{t+j+1}\right) \tag{6.4.18}$$

$$b_t = \sum_{j=0}^{\infty}\beta^j E_t y_{t+j} - \frac{1}{1-\beta}c_t. \tag{6.4.19}$$

Equation (6.4.19) asserts that a household's debt due at $t$ equals the expected present value of its endowment minus the expected present value of its consumption stream. A high debt thus indicates a large expected present value of 'surpluses' $y_t - c_t$.

Recalling the form of the endowment process (6.4.3), we can compute

$$E_t\sum_{j=0}^{\infty}\beta^j z_{t+j} = \left(I-\beta\check{A}\right)^{-1} z_t \tag{6.4.20a}$$

$$E_{t+1}\sum_{j=0}^{\infty}\beta^j z_{t+j+1} = \left(I-\beta\check{A}\right)^{-1} z_{t+1} \tag{6.4.20b}$$

$$E_t\sum_{j=0}^{\infty}\beta^j z_{t+j+1} = \left(I-\beta\check{A}\right)^{-1} \check{A}z_t. \tag{6.4.20c}$$

Substituting these formulas into (6.4.18) and (6.4.19) and using (6.4.3$a$) gives the following representation for the consumer's optimum decision rule:[11]

$$c_{t+1} = c_t + (1-\beta)\check{G}\left(I-\beta\check{A}\right)^{-1}\check{C}w_{t+1} \tag{6.4.21a}$$

$$b_t = \check{G}\left(I-\beta\check{A}\right)^{-1} z_t - \frac{1}{1-\beta}c_t \tag{6.4.21b}$$

$$y_t = \check{G}z_t \tag{6.4.21c}$$

$$z_{t+1} = \check{A}z_t + \check{C}w_{t+1} \tag{6.4.21d}$$

Representation (6.4.21) reveals several things about the optimal decision rule. (1) The *state* consists of the endogenous part $c_t$ and the exogenous part

[11] See appendix A of chapter 18 for a reinterpretation of precisely these outcomes in terms of a competitive equilibrium of a model with a complete set of markets in history- and date-contingent claims to consumption.

$z_t$.[12] These contain all information that is relevant for forecasting future $c, y, b$. Notice that financial assets $b_t$ have disappeared as a component of the state because they are properly encoded in $c_t$. (2) According to (6.4.21), consumption is a random walk with innovation $(1-\beta)d(\beta)w_{t+1}$ as implied also by (6.4.17). This outcome confirms that the Euler equation (6.4.6) is built into the solution. That consumption is a random walk of course implies that it does not possess an asymptotic stationary distribution, at least so long as $z_t$ exhibits perpetual random fluctuations, as it will generally under (6.4.3).[13] This feature is inherited partly from the assumption that $\beta R = 1$. (3) The impulse response function of $c_t$ is a "box": for all $j \geq 1$, the response of $c_{t+j}$ to an increase in the innovation $w_{t+1}$ is $(1-\beta)d(\beta) = (1-\beta)\check{G}(I-\beta\check{A})^{-1}\check{C}$. (4) Solution (6.4.21) reveals that the joint process $c_t, b_t$ possesses the property that Granger and Engle (1987) called *cointegration*. In particular, *both* $c_t$ and $b_t$ are non-stationary because they have unit roots (see representation (6.4.13) for $b_t$), but there is a linear combination of $c_t, b_t$ that *is* stationary provided that $z_t$ is stationary. From (6.4.19), the linear combination is $(1-\beta)b_t + c_t$. Accordingly, Granger and Engle would call $[\,(1-\beta) \quad 1\,]$ a cointegrating vector that, when applied to the nonstationary vector process $[\,b_t \quad c_t\,]'$, yields a process that is asymptotically stationary. Equation (6.4.9) can be arranged to take the form

$$(1-\beta)\, b_t + c_t = (1-\beta)\, E_t \sum_{j=0}^{\infty} \beta^j y_{t+j}, \tag{6.4.22}$$

which asserts that the 'cointegrating residual' on the left side equals the conditional expectation of the weighted sum of future incomes on the right.[14]

---

[12] Something subtle is going on in system (6.4.18)-(6.4.19): the *state* variable $c_t$ is actually responding to the shocks $w_t$ to $z_t$ that shape time $t$ nonfinancial income $y_t$ while the "outcome" variable $b_t$ has actually been chosen at $t-1$ and so it is predetermined. It can be expressed as an exact function of the time $t$ "state" variables $c_t, z_t$ only because $c_t$ adjusts to $z_t$ in just the right way.

[13] The failure of consumption to converge will occur again in chapter 18 when we drop quadratic utility and assume that consumption must be nonnegative.

[14] See Campbell and Shiller (1988) and Lettau and Ludvigson (2001, 2004) for interesting applications of related ideas.

### 6.4.2. Debt dynamics

If we subtract equation (6.4.21$b$) evaluated at time $t$ from equation (6.4.21$b$) evaluated at time $t+1$, we obtain

$$b_{t+1} - b_t = \check{G}\left(I - \beta\check{A}\right)^{-1}(z_{t+1} - z_t) - \frac{1}{1-\beta}(c_{t+1} - c_t).$$

Substituting $z_{t+1} - z_t = (\check{A} - I)z_t + \check{C}w_{t+1}$ and equation (6.4.21$a$) into the above equation and rearranging gives

$$b_{t+1} - b_t = \check{G}\left(I - \beta\check{A}\right)^{-1}\left(\check{A} - I\right)z_t. \tag{6.4.23}$$

This equation indicates how $b_{t+1}$ is predetermined at time $t$ as a function of $b_t$ and $z_t$.

Representation (6.4.23) for the consumer's optimal borrowing rule can be solved backwards to express $b_t$ as a function of $z^{t-1} = [z_{t-1}, z_{t-2}, \ldots, z_0]$ and initial condition $b_0$. Because $b_t$ depends on the entire history in a non-trivial way, we say that it is *history dependent*, as is consumption, with which debt is cointegrated, as we have observed above. History dependence of optimal debt and consumption in the LQ consumption smoothing model is a tell-tale sign of markets being incomplete in a sense that we'll investigate in section 6.6.

### 6.4.3. Two classic examples

We illustrate formulas (6.4.21) with the following two examples. In both examples, the endowment follows the process $y_t = z_{1t} + z_{2t}$ where

$$\begin{bmatrix} z_{1t+1} \\ z_{2t+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} z_{1t} \\ z_{2t} \end{bmatrix} + \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}\begin{bmatrix} w_{1t+1} \\ w_{2t+1} \end{bmatrix} \tag{6.4.24}$$

where $w_{t+1}$ is an i.i.d. $2 \times 1$ process distributed as $\mathcal{N}(0, I)$. Here $z_{1t}$ is a permanent component of $y_t$ while $z_{2t}$ is a purely transitory component.

**Example 1.** Assume that the consumer observes the state $z_t$ at time $t$. This implies that the consumer can construct $w_{t+1}$ from observations of $z_{t+1}$ and $z_t$. Application of formulas (6.4.21) implies that

$$c_{t+1} - c_t = \sigma_1 w_{1t+1} + (1-\beta)\,\sigma_2 w_{2t+1}. \tag{6.4.25}$$

Since $1-\beta = \frac{r}{1+r}$ where $R = (1+r)$, formula (6.4.25) shows how an increment $\sigma_1 w_{1t+1}$ to the permanent component of income $z_{1t+1}$ leads to a permanent one-for-one increase in consumption and no increase in savings $-b_{t+1}$; but how the purely transitory component of income $\sigma_2 w_{2t+1}$ leads to a permanent increment in consumption by a fraction $(1-\beta)$ of transitory income, while the remaining fraction $\beta$ is saved, leading to a permanent increment in savings $-b_{t+1}$. Application of formula (6.4.23) to this example shows that

$$b_{t+1} - b_t = -z_{2t} = -\sigma_2 w_{2t}, \tag{6.4.26}$$

which confirms that none of $\sigma_1 w_{1t}$ is saved, while all of $\sigma_2 w_{2t}$ is saved.

**Example 2.** We build on Muth's classic reverse engineering exercise that we described in section 6.2. Thus, we assume that the income process $\{y_t\}$ obeys state space system (6.2.2) and that the consumer observes $y_t$, and its history up to $t$, but not the hidden state $x_t$ at time $t$. In terms of the notation used in system (6.4.24), this amounts to assuming that the consumer observes $y_t = z_{1t} + z_{2t}$ but not $z_{1t}$ and $z_{2t}$ separately. Under this assumption, it is appropriate to use an *innovation representation* to form $\check{A}, \check{C}, \check{G}$ in formulas (6.4.21). In particular, using the innovations representation (6.2.7) derived in subsection 6.2, the appropriate state-space representation for $y_t$ is

$$\begin{bmatrix} y_{t+1} \\ a_{t+1} \end{bmatrix} = \begin{bmatrix} 1 & -(1-K) \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ a_t \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} a_{t+1}$$
$$y_t = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_t \\ a_t \end{bmatrix}$$

where $K$ is the Kalman gain and $a_t = y_t - E[y_t | y^{t-1}]$. From subsection 6.2, we know that $K \in [0,1]$ and that $K$ increases as $\frac{\sigma_1^2}{\sigma_2^2}$ increases, i.e., as the ratio of the variance of the permanent shock to the variance of the transitory shock to income increases. Applying formulas (6.4.21) implies

$$c_{t+1} - c_t = [1 - \beta(1-K)]\, a_{t+1} \tag{6.4.27}$$

where the endowment process can now be represented in terms of the univariate innovation to $y_t$ as

$$y_{t+1} - y_t = a_{t+1} - (1-K)\, a_t. \tag{6.4.28}$$

Equation (6.4.28) indicates that the consumer regards a fraction $K$ of an innovation $a_{t+1}$ to $y_{t+1}$ as *permanent* and a fraction $1-K$ as purely transitory. He

permanently increases his consumption by the full amount of his estimate of the permanent part of $a_{t+1}$, but by only $(1-\beta)$ times his estimate of the purely transitory part of $a_{t+1}$. Therefore, in total he permanently increments his consumption by a fraction $K + (1-\beta)(1-K) = 1 - \beta(1-K)$ of $a_{t+1}$ and saves the remaining fraction $\beta(1-K)$ of $a_{t+1}$. According to equation (6.4.28), the first difference of income is a first-order moving average, while (6.4.27) asserts that the first difference of consumption is i.i.d. Application of formula (6.4.23) to this example shows that

$$b_{t+1} - b_t = (K-1)\, a_t, \tag{6.4.29}$$

which indicates how the fraction $K$ of the innovation to $y_t$ that is regarded as permanent influences the fraction of the innovation that is saved.

### *6.4.4. Spreading consumption cross sections*

The common unit root affecting $c_t, b_t$ causes the time $t$ variance of $c_t$ to grow linearly with $t$. If we think of the initial distribution as describing the joint distribution of $c_0, b_0$ for a cross section of *ex ante* identical households born at time $0$, then our formulas describe the evolution of cross section distributions of $b_t, c_t$ as the population of households ages. Cross-section consumption distributions spreads out as time passes.[15] [16] We'll reconsider this outcome in the next several sections of this chapter, and also in chapters 9, 18, 19, 22 and 23.

[15] See Deaton and Paxton (1994) and Storesletten, Telmer, and Yaron (2004) for evidence that cross section distributions of consumption spread out with age.

[16] **Tom, at some point we have to acknowledge the limitation of our solution to the LQ savings problem under the assumption of only risk-free borrowing and lending.** Yes, we are making sure that agents' first-order conditions and the boundary condition (6.4.5) are satisfied, but we ignore second-order conditions when letting consumption drift above the bliss point in our quadratic utility specification. Since utility decreases for any consumption in excess of the bliss level, agents would never want to consume anything above that bliss level. Do you have any suggestion how we best can convey this limitation of our analysis?

## 6.5. A "borrowers and lenders" closed economy

Up to now, we have set $R = \beta^{-1}$ and taken it to be have determined "outside" the model. Saying that we are analyzing a "small open economy" is a common justification for this assumption. By adapting ideas of Truman Bewley (1977, 1980) to the present setting, there is a way to fill out the consumer's environment that makes $R = \beta^{-1}$ an equilibrium outcome. To do this, we imagine that our consumer is one of a continuum of unit measure of *ex ante* identical, but *ex post* heterogeneous consumers who trade a single risk-free one-period bond whose price is $\beta$. Let individual consumers be indexed by $i \in [0,1]$ and we assume that their initial financial wealth are all equal to zero, $b_0^i = 0$ for all $i$. The consumers are *ex ante* identical in their preferences and in their stochastic processes for their non-financial incomes as given by (6.4.3). But they are *ex post* heterogeneous in realizations of their non-financial incomes from time 0 on. Specifically, we assume that their initial vectors $z_0$ are independent draws from the stationary distribution of their common statistical model (6.4.3), let $z_0^i$ denote the initial vector of consumer $i$ with an associated initial non-financial income of $y_0^i = \check{G} z_0^i$. Future realizations of non-financial income are driven by consumers' idiosyncratic draws of the vector of shocks $w_{t+1}$, let $w_{t+1}^i$ denote consumer $i$'s vector of shocks. Our assumption about the initial distribution of the vectors $z_0^i$ implies that the cross-section average of non-financial income equals the mean of its stationary distribution each period. Since we are now analyzing a closed economy, it follows that the cross-section mean of consumption equals the stationary mean of non-financial income each period.[17] But what about higher moments of the cross-section distribution of consumption?

With the intitial conditions that we have specified, each consumer's consumption follows a random walk and his/her financial wealth is co-integrated with his/her own consumption. But average debt in the cross-section of all consumers stays at zero. Each period, the market for one-period risk-free loans clears at the same constant gross interest rate $R = \beta^{-1}$. At this interest rate, appropriate weighted sums of one-period risk-free borrowing equal appropriate weighted sums of one-period risk free lending at each $t \geq 0$.

Thus, we have specified an economy in which a set of people are borrowing and lending with each other at a gross risk-free interest rate of $R = \beta^{-1}$.

[17] The cross-section mean of consumption appropriately accounts for any negative consumption rates as being production.

Across all consumers, risk-free loans are in zero net supply. We have arranged primitives so that $R = \beta^{-1}$ clears the market for risk-free loans at zero net supply each period. So the risk-free loans are being made from one consumer to another within our closed set of agents. In this way, we have dispensed with the "small open economy" assumption and created our first "Bewley model". We'll encounter many more in chapter 19.

### *6.5.1. An inefficiency*

Given initial condition $b_0^i = 0$ and using expressions (6.4.10) and (6.4.12), agent $i$'s consumption in the first period is

$$c_0^i = (1-\beta)\sum_{j=0}^{\infty}\beta^j E_0 y_j^i = (1-\beta)\check{G}\left(I-\beta\check{A}\right)^{-1} z_0^i. \tag{6.5.1}$$

From equation (6.4.21$a$), we know that

$$c_{t+1}^i = c_t^i + h w_{t+1}^i,$$

where $h = (1-\beta)\check{G}(I-\beta\check{A})^{-1}\check{C}$ so that

$$\begin{aligned} E_0 c_t^i &= c_0^i \\ E_0\left(c_t^i - c_0^i\right)^2 &= thh'. \end{aligned} \tag{6.5.2}$$

Because each consumer $i$ dislikes variance in consumption, he would prefer a completely smoothed consumption stream in which $c_t^i = c_0^i$ for all $t \geq 0$. Furthermore, such an allocation would be physically feasible since the cross section average of our continuum of consumers actually equals the mean of the stationary distribution of non-financial income. If at time 0 everyone could get together and agree to share society's non-financial income in that manner, each person would prefer that allocation to the risky allocation they attain in our closed economy in which each period they trade one-period risk-free bonds. In the next section, we describe an environment that our consumers would prefer to find themselves living in, one with the same physical opportunities but with additional markets. In particular, we add enough securities to the economy so that, by trading them, our consumers can completely insure themselves against random fluctuations in their non-financial incomes.

I have deleted any opportunity to trade before the economy starts at time 0.

The entire section 6.6 should be marked yellow (followed by the earlier version of 6.6).

## 6.6. Consumption smoothing with complete markets

It is instructive to consider a complete markets version of the LQ permanent income model studied in sections 6.4 and 6.5. Specifically, the consumers are provided with the opportunity to avoid all fluctuations in consumption by trading state-contingent claims to future consumption. As in section 6.4 we initially suppress the index $i$ for a particular agent and simply study a general problem solved by every consumer. Earlier budget constraint (6.4.2) is now replaced by

$$c_t + b_{t-1}(z_t) = \int q(z_{t+1}|z_t)\, b_t(z_{t+1})\, dz_{t+1} + y_t, \tag{6.6.1}$$

where $q(z_{t+1}|z_t)$ is a kernel for pricing bundles of one-period state contingent claims and $b_t(z_{t+1})$ is a function describing a bundle of state-contingent debts due at time $t+1$ that are chosen at time $t$. That is, function $b_t(z_{t+1})$ describes the consumer's financial portfolio, chosen at time $t$ of one-period state-contingent securities that promises to pay off $b_t(z_{t+1})$ units of time $t+1$ consumption at time $t+1$ in Markov state $z_{t+1}$. The function $b_t(z_{t+1}) : I\!R \to I\!R$ is measurable at time $t$ and maps next period's $z_{t+1} \in I\!R$ into a $z_{t+1}$-contingent payment at time $t+1$.

Following our approach in section 6.4 of arbitrarily setting $R^{-1} = \beta$ (and then showing in section 6.5 that this price is an equilibrium outcome), we guess that the equilibrium pricing kernel will be

$$q(z_{t+1}|z_t) = \beta\phi(z_{t+1}|z_t), \tag{6.6.2}$$

where $\phi(z_{t+1}|z_t)$ is the probability density of $z_{t+1}$ conditional on $z_t$. Pricing kernel (6.6.2) discounts state-contingent claims on consumption at date $t+1$ by the product of the time discount factor $\beta$ and conditional probabilities of the events at which claims pay off. Sometimes the kernel $q(z_{t+1}|z_t)$ is called a state-price density, and the function $b_t(z_{t+1})$ is said to describe a bundle of one-period ahead Arrow securities.

With the pricing kernel in hand, we can price "redundant securities" that are simply bundles of state-contingent claims.[18] For example, in state $z_t$ at

[18] For example, it is redundant to add a risk-free one-period bond to the state-contingent one period securities that the consumer can trade in our complete markets model. If that redundant asset were added, a no-arbitrage profits argument like one that we shall apply repeatedly in chapters 9, 14, and 15 would imply formula (6.6.15).

time $t$, the price of a claim to one unit of consumption at time $t+1$ when realization $z_{t+1} \in A \subset I\!R$ is

$$\int_A q\left(z_{t+1}|z_t\right) \cdot 1\, dz_{t+1}.$$

Another example is a risk-free one period bond sold at time $t$ that pays one unit of consumption for sure (i.e., for all $z_{t+1}$) at time $t+1$. Pricing kernel (6.6.2) implies that the price of such a one-period risk-free bond is

$$\int q\left(z_{t+1}|z_t\right) dz_{t+1} = \beta \int \phi\left(z_{t+1}|z_t\right) dz_{t+1} = \beta. \tag{6.6.3}$$

Thus, the gross interest rate on one-period risk free loans is $R = \beta^{-1}$. This outcome indicates a sense in which our guess of pricing kernel (6.6.2) might correspond to its "incomplete markets" counterpart in section 6.4 model.

We conjecture that the equilibrium allocation is a time invariant consumption level,

$$c_t = \bar{c} = c_0, \quad \forall t \geq 0, \tag{6.6.4}$$

where $c_0$ is given by equation (6.5.1), i.e., the consumer's first-period consumption in the incomplete markets version of the economy. To construct a financial plan that supports this time invariant consumption level, we revisit consumption function (6.4.12) in the incomplete markets economy. In that economy recall that the consumer enters period $t$ in Markov state $z_t$ with non-contingent debt $b_t$ and chooses a consumption level $c_t$ such that the conditional expectations of all future consumption satisfy $E_t c_{t+j} = c_t$ for all $j > 0$, which reflect first-order necessary conditions in the consumer's optimization problem. Our suggested analogue in the present complete markets economy is that the consumer enters period $t$ with a state-contingent debt $b_{t-1}(z_t)$ such that our conjectured constant consumption level $\bar{c}$ can be afforded. Thus, substitute (6.6.4) into (6.4.12) and solve for the following state-contingent debt:

$$b_{t-1}\left(z_t\right) = \check{G}\left(I - \beta \check{A}\right)^{-1} z_t - \frac{1}{1-\beta}\bar{c} \equiv b\left(z_t, \bar{c}\right). \tag{6.6.5}$$

As this conjectured debt function $b(z_t, \bar{c})$ will be confirmed below it is rather remarkable that indebtedness when entering period $t$ does not depend on history

but only on the realization of $z_t$.[19] Actually, it is the hallmark of risk sharing in a complete markets equilibrium that idiosyncratic risks are not born by anyone.

We now confirm that the conjectured portfolio choice attains the constant consumption level $\bar{c}$ and is affordable to the consumer. Substituting conjectures (6.6.2) and (6.6.5) into budget constraint (6.6.1) and using formulas (6.4.20$a$), (6.4.20$b$) and (6.4.3$b$) gives the following expression

$$\begin{aligned} c_t + & E\left[\sum_{j=0}^{\infty} \beta^j y_{t+j} \middle| z_t\right] - \frac{1}{1-\beta}\bar{c} \\ &= \int \beta\phi\left(z_{t+1}|z_t\right)\left(E\left[\sum_{j=0}^{\infty} \beta^j y_{t+j+1} \middle| z_{t+1}\right] dz_{t+1} - \frac{1}{1-\beta}\bar{c}\right) + y_t \\ &= E\left[\sum_{j=0}^{\infty} \beta^j y_{t+j} \middle| z_t\right] - \frac{\beta}{1-\beta}\bar{c}\,, \end{aligned}$$

where the last equality invokes the law of iterated expectations on the right side of the budget constraint. After a cancellation of terms and a collection of others, this expression for the budget constraint simplifies to $c_t = \bar{c}$. Thus, we have confirmed that all one-period budget constraints at dates $t \geq 1$ and for all paths of $\{z_t\}_t$ yield the same constant consumption level $\bar{c}$. Though, it remains to be shown that the initial budget constraint at time 0 also is satisfied, i.e., whether or not the consumption level $\bar{c}$ and portfolio bundle $b(z_1, \bar{c})$ are affordable. We do this by manipulating expression (6.5.1) in such a way that the budget constraint at time 0 emerges under our conjecture that $\bar{c} = c_0$ in the incomplete markets economy:

$$\begin{aligned} \frac{\bar{c}}{1-\beta} &= E\left[\sum_{j=1}^{\infty} \beta^j y_j \middle| z_0\right] + y_0\,, \\ \bar{c} + \frac{\beta}{1-\beta}\bar{c} &= \int \beta\phi\left(z_1|z_0\right) E\left[\sum_{j=1}^{\infty} \beta^{j-1} y_j \middle| z_1\right] dz_1 + y_0\,, \end{aligned}$$

[19] Financial wealth in all periods depends also on the second argument $\bar{c}$ in $b(z_t, \bar{c})$, which reflects the initial value of a consumer's endowment process at time 0. In our context, each consumer randomly draws an initial vector $z_0$ as described in section 6.5, which determines the initial value of a consumer's endowment process as of time 0.

$$\bar{c} = \int \beta\phi\left(z_1|z_0\right)\left(\check{G}\left(I-\beta\check{A}\right)^{-1} z_1 - \frac{1}{1-\beta}\bar{c}\right) dz_1 + y_0\,,$$
$$\bar{c} = \int q\left(z_1|z_0\right) b\left(z_1,\bar{c}\right) dz_1 + y_0\,,$$

where we once again have used the law of iterated expectations in the transition between the first two equations, and the last equation is the consumer's budget constraint at time $0$. We conclude that the constant consumption level $\bar{c}$ is indeed equal to the first-period consumption $c_0$ in the incomplete markets economy, and the portfolio choice $b(z_1,\bar{c})$ is affordable at time $0$.

Finally, we adopt the assumption of section 6.5 that in the economy's first period $0$, consumers' initial $z_0^i$ are independent draws from the stationary distribution of their common statistical model (6.4.3), where individual consumers are indexed by $i \in [0,1]$. Under that assumption, together with the initial conditions of no outstanding debts at the beginning of period $0$, this present section has characterized the optimization behavior of individual consumers in a general equilibrium of the complete markets economy. The individual-specific draws of $z_0^i$ map into different constant consumption levels $\bar{c}^i = c_0^i$ as given by equation (6.5.1). Instead of consumption spreading out over time as in the incomplete markets economy, each consumer is fixated in an unchanging initial distribution of consumption inequality.

A lucky draw of $z_0^i$ at the beginning of period $0$ manifests both as high consumption $\bar{c}^i$ in period $0$ and all future periods as given by (6.5.1), as well as and not surprisingly, lower indebtedness $b(z_t^i,\bar{c})$ in all future Markov states $z_t^i$ as given by (6.6.5). Evidently, the effect on indebtedness is an equal downward shift in $b(z_t^i,\bar{c})$ across all possible Markov states $z_t^i$, which provides the financial conditions for affording a higher time invariant consumption level $\bar{c}^i$. Regarding differences in $b(z_t^i,\bar{c})$ across states $z_t^i$, they serve to insure against idiosyncratic shocks $w_t^i$ to consumer $i$'s endowment process, i.e., the source of his/her non-financial income. Besides heterogeneity in the realizations of those idiosyncratic shocks $w_t^i$ and the randomly drawn initial values of $z_0^i$, all consumers share the same parametric specification of their endowment processes as given by (6.4.3). It follows that the optimum decision rule for indebtedness, $b(\cdot,\cdot)$, is the same for all consumers and hence, bears no index $i$.

The composition of consumer $i$'s optimum portfolio always stays the same: the same bundle $b(z_{t+1}^i,\bar{c}^i)$ is purchased at every date $t \geq 0$ regardless of the present Markov state $z_t^i$. However, the purchase price of that portfolio depends

on the pricing kernel $q(z_{t+1}^i|z_t^i)$ which is a function of the Markov state $z_t^i$ at the time of purchase. As confirmed above, this state-dependent purchase price of the optimum portfolio,

$$\int q\left(z_{t+1}^i|z_t^i\right) b\left(z_{t+1}^i, \bar{c}^i\right) dz_{t+1}^i\,,$$

is affordable to each consumer $i$ in every period $t$ and any Markov state $z_t^i$.

From a welfare perspective all consumers with their respective initial draws of $z_0^i$ attain a higher expected utility under complete markets by consuming $\bar{c}^i$ in every period. As compared to their stochastic consumption in the incomplete markets economy when consumer $i$'s time 0 expectations of his/her future consumption are equal to $E_0 c_t^i = c_0^i = \bar{c}^i$ at all forecast horizons $t$, i.e., the means are the same as the constant consumption level $\bar{c}^i$ that they would enjoy in the complete markets economy. It follows that the consumers in the incomplete markets economy have a lower expected utility since their consumption varies around a mean level equal to the time invariant consumption in the complete markets economy, and that variance of consumption even grows with the forecast horizon as shown in (6.5.2). Consumers dislike variance in consumption because of their concave utility function.

We shall study more general complete markets models of consumption smoothing in chapter 9 and more general incomplete markets models in chapter 19. In those chapters, interest rates and state-contingent prices are among the objects to be determined within an equilibrium.

Here ends the yellow markings of the entire section 6.6, followed by the earlier version of section 6.6.

========== EARLIER VERSION BELOW

It is instructive to take the LQ permanent income model studied in section 6.4 and, by altering assumptions about *what* is determined *when* in equation (6.4.7), provide the consumer with the opportunity to avoid all fluctuations in consumption by trading state-contingent claims to future consumption. To start, return to equations (6.4.6) and (6.4.7) that we repeat for convenience:

$$E_t c_{t+1} = c_t. \tag{6.6.6}$$

$$b_t = \sum_{j=0}^{\infty} \beta^j \left(y_{t+j} - c_{t+j}\right). \tag{6.6.7}$$

Recall that (6.6.6) is an Euler equation for consumption, i.e., a first-order necessary condition for constrained optimization, and that (6.6.7) is the outcome

of solving in the forward direction the first-order linear difference equation that is the consumer's one-period budget constraint using the method described in appendix B of chapter 2. As we did in section 6.4, take conditional expectations on both sides of (6.6.7) and use (6.6.6) and the law of iterated expectations to deduce

$$b_t = \sum_{j=0}^{\infty} \beta^j E_t y_{t+j} - \frac{1}{1-\beta} c_t. \tag{6.6.8}$$

As in section 6.4, we assume that non-financial income $y_t$ is governed by the state-space representation (6.4.3), namely,

$$\begin{aligned} z_{t+1} &= \check{A} z_t + \check{C} w_{t+1} \\ y_t &= \check{G} z_t, \end{aligned}$$

where $w_{t+1}$ is an i.i.d. process with mean zero and identity contemporaneous covariance matrix, all eigenvalues of $\check{A}$ are strictly less than $\frac{1}{\sqrt{\beta}}$ in magnitude, and $\check{G}$ is a vector that sets $y$ to be a linear combination of $z_t$. Computing conditional expectations of $y_{t+j}, j \geq 0$ and substituting into (6.6.8), we obtain

$$c_t = (1-\beta) \left[ \check{G} \left( I - \beta \check{A} \right)^{-1} z_t - b_t \right]. \tag{6.6.9}$$

Stare at equation (6.6.9) and think about what is predetermined at $t$ and what is free to adjust to shocks to $z_t$, i.e., what are *states* and what are *outcomes* at $t$. In section 6.4, we regarded both $b_t$ and $z_t$ as components of the consumer's state at time $t$, each having been determined before time $t$, with the scalar $-b_t$ being the consumer's financial assets in the form of risk-free claims to time $t$ consumption that the consumer purchased at time $t-1$ and the vector $z_t$ being information that is useful for predicting future non-financial incomes $\{y_{t+j}\}_{j=0}^{\infty}$. In section 6.4, we interpreted (6.6.9) as the consumer's optimal decision rule for $c_t$ as a function of these state variables, i.e., the consumer's "consumption function".

By making different assumptions about what variables in (6.6.9) are set before time $t$ (they are "predetermined" at time $t$) and what variables can be adjusted at time $t$, we can allow the consumer to buy and sell consumption insurance. To show this, we now construct a consumption plan that satisfies both conditions (6.6.6) and (6.6.7) and also a special case of equation (6.6.9) in which

$$c_t = \bar{c} = c_0, \quad \forall t \geq 0 \tag{6.6.10}$$

where $c_0$ is given by equation (6.4.12) for the optimal time 0 consumption in the section 6.4 consumption-smoothing model. Because it completely smooths consumption and has the same time 0 consumption, this $c_t = \bar{c}$ plan awards higher discounted utility to the consumer than does the random walk consumption plan implied by decision rule (6.6.9), an assertion that we'll soon verify.

To construct a financial plan that supports constant consumption $c_t = c_0$, we let the consumer trade one-period state-contingent securities. (These are examples of the Arrow securities to be studied in chapter 9.) Temporarily take $\bar{c}$ as given, substitute (6.6.10) into (6.6.9) and solve for the following state-dependent debt function $b(z_t, \bar{c})$:

$$b_{t-1}(z_t, \bar{c}) = \check{G}\left(I - \beta\check{A}\right)^{-1} z_t - \frac{1}{1-\beta}\bar{c}. \tag{6.6.11}$$

The first term on the right side is the present value of non-financial income in state $z_t$, while the second term is minus the present value of consumption. The function $b_{t-1}(z_t, \bar{c})$ on the left side of the equation describes the consumer's portfolio, chosen at time $t-1$ in state $z_{t-1}$, of one-period state-contingent securities that at time $t-1$ had promised to pay off one unit of time $t$ consumption at time $t$ in Markov state $z_t$. The function $b_{t-1}(z_t) : I\!R \to I\!R$ maps $z_{t-1} \in I\!R$ into a $z_t$-contingent payment at time $t+1$. As part of our reverse engineering exercise, we assume that equation (6.6.11) describes the consumer's optimal portfolio. To pin down the constant $\bar{c}$, we assume that the consumer enters time 0 owing debt $\bar{b}_0$ and that $\bar{c}$ solves the following version of (6.6.11) at time $t = 0$:

$$\bar{c} = (1-\beta)\left[\check{G}\left(I - \beta\check{A}\right)^{-1} z_0 - \bar{b}_0\right], \tag{6.6.12}$$

so that $b(z_0, \bar{c}) = \bar{b}_0$ also satisfies equation (6.6.11).

Evidently model (6.6.11)-(6.6.12) has extinguished the history dependence of both debt and consumption exhibited by the "incomplete markets" LQ permanent income consumption smoothing model of section 6.4. Consumption given by decision rule (6.6.12) is obviously not history dependent, while the debt decision given by (6.6.11) depends on Markov state $z_t$ but not earlier Markov states, $z_{t-1}, z_{t-2}, \ldots$. To understand these outcomes, we now describe how we have backed ourselves into a model in which each period the consumer trades a complete list of one-period state-contingent securities that pay off amounts next period that depend on next period's information vector $z$. To make this work with a consumption rate that is constant across time and Markov states $z$, we

must reverse engineer prices of those securities that they induce the consumer to want to smooth consumption completely across both states and time by setting $c_t = \bar{c} = c_0$. We can construct prices of state-contingent securities that convince the consumer to do that. Such prices will turn out to imply that the price at time $t$ of one unit of consumption for sure at time $t+1$ is $\beta$ for all $t \geq 0$, just as it is in the section 6.4 model in which the consumer can trade only a risk-free one-period bond.

Our consumer now faces a sequence of one-period budget constraints whose time $t \geq 0$ component is

$$c_t + b_{t-1}(z_t) = \int q(z_{t+1}|z_t)\, b_t(z_{t+1})\, dz_{t+1} + y_t \tag{6.6.13}$$

where $q(z_{t+1}|z_t)$ is a kernel for pricing bundles of one-period state contingent claims and a function $b_t(z_{t+1})$ describes a bundle of state-contingent debts due at time $t+1$ that are chosen at time $t$. As above, a function $b_t(z_{t+1}) : I\!R \to I\!R$ maps $z_t \in I\!R$ into a $z_{t+1}$-contingent payment of time $t+1$ consumption goods. As the next step in our reverse engineering exercise, we arbitraily set the pricing kernel to

$$q(z_{t+1}|z_t) = \beta\phi(z_{t+1}|z_t), \tag{6.6.14}$$

where $\phi(z_{t+1}|z_t)$ is the probability density of $z_{t+1}$ conditional on $z_t$. For now, the pricing kernel is set outside the model, just as earlier we took a one-period gross risk-free interest rate $R = \beta^{-1}$ to have been determined outside the section 6.4 consumption-smoothing model. (Soon we'll describe a setting in which this pricing kernel turns out to be an equilibrium object that is determined inside an augmented version of the model.) Pricing kernel (6.6.14) discounts state-contingent claims on consumption at date $t+1$ by the product of the time discount factor $\beta$ and conditional probabilities of the events at which claims pay off. Sometimes the kernel $q(z_{t+1}|z_t)$ is called a state-price density, and the function $b_t(z_{t+1})$ is said to describe a bundle of one-period ahead Arrow securities.

With the pricing kernel in hand, we can price "redundant securities" that are simply bundles of state-contingent claims.[20] An Arrow security issued at

[20] For example, it is redundant to add a risk-free one-period bond to the state-contingent one period securities that the consumer can trade in our complete markets model. If that redundant asset were added, a no-arbitrage profits argument like one that we shall apply repeatedly in chapters 9, 14, and 15 would imply formula (6.6.3).

time $t$, Markov state $z_t$ that pays off 1 unit of consumption at time $t+1$ whenever the realized time $t$ state $z_{t+1}$ lies in interval $(z_{t+1}, z_{t+1} + dz_{t+1})$ is priced at time $t$ by $dq(z_{t+1}|z_t)$. Thus, in state $z_t$ at time $t$, the price of a claim to one unit of consumption at time $t+1$ when realization $z_{t+1} \in A \subset I\!R$ is

$$\int_A q\left(z_{t+1}|z_t\right) 1 dz_{t+1}.$$

For example, a risk-free one period bond sold at time $t$ pays one unit of consumption for sure (i.e., for all $z_{t+1}$) at time $t+1$. Pricing kernel (6.6.14) implies that the price of such a one-period risk-free bond is

$$\int q\left(z_{t+1}|z_t\right) dz_{t+1} = \beta \int \phi\left(z_{t+1}|z_t\right) dz_{t+1} = \beta. \tag{6.6.15}$$

Thus, the gross interest rate on one-period risk free loans is $\beta^{-1}$. This outcome indicates a sense in which our reverse engineering exercise here corresponds to its "incomplete markets" counterpart in the section 6.4 model.

With pricing kernel (6.6.14), the consumer's time $t$ one-period budget constraint (6.6.13) at $(t, z_t)$ becomes

$$c_t + b_{t-1}\left(z_t\right) = \beta E b_t\left(z_{t+1}\right)|z_t + y_t \tag{6.6.16}$$

instead of the section 6.4 budget constraint

$$c_t + b_t = \beta b_{t+1} + y_t \tag{6.6.17}$$

in which $b_{t+1}$ is a scalar that the consumer chooses at time $t$ and $b_{t-1}$ is a scalar that the consumer chose at time $t-1$ as a function of information $z^{t-1} = [z_{t-1}, z_{t-2}, \ldots, z_0]$.

Notice that $\bar{c}$ in equation (6.6.12) equals $c_0$ given by the optimal decision rule (6.4.12) for the section 6.4 incomplete markets version of the LQ consumption smoothing model.[21] From equation (6.4.21$a$), we know that in the incomplete markets version of the model

$$c_{t+1} = c_t + h w_{t+1},$$

[21] It is an incomplete markets model in the sense that the consumer can trade only risk-free one-period securities instead of the complete set of securities present in the model of this section.

where $h = (1-\beta)\check{G}(I-\beta\check{A})^{-1}\check{C}$ and equations (6.5.2) that depict spreading cross-section distributions of consumption again prevail. Because the consumer dislikes variance in consumption, he prefers the completely smoothed consumption stream that he chooses in the complete markets version of the model to the random walk consumption stream that he chooses in the section 6.5 risk-free-bond-only economy.

We shall study more general complete markets models of consumption smoothing in chapter 9 and more general incomplete markets models in chapter 19. In those chapters, interest rates and state-contingent prices are among the objects to be determined within an equilibrium.

## 6.7. The Lucas Critique

We can use our chapter 2 tools to describe reasoning that led Robert E. Lucas, Jr., (1976) to criticize some procedures that were then widely used in applied work, and that occasionally still are. Assume a jointly stationary Gaussian stochastic process for $\{x_t, y_t, d_t\}_{t=0}^{\infty}$ generated by a state-space system

$$x_{t+1} = Ax_t + Cw_{t+1} \tag{6.7.1a}$$

$$y_t = Gx_t \tag{6.7.1b}$$

$$d_t = Sx_t \tag{6.7.1c}$$

$$S = HG\,(I-\delta A)^{-1} \tag{6.7.1d}$$

where $\delta \in (0,1)$ is a discount factor and $A$ is a stable matrix, and $x_0 \sim \mathcal{N}(\mu, \Sigma)$, where $\mu, \Sigma$ are the mean vector and covariance matrix, respectively, of a stationary distribution for the $\{x_t\}_{t=0}^{\infty}$ process.[22] As usual, $w_{t+1}$ is the time $t+1$ component of an i.i.d. sequence of $\mathcal{N}(0, I)$ random vectors. Here $d_t$ is a vector of decisions that an econometrician observes. The resolvent operator $(I-\delta A)^{-1}$ appears on the right side of (6.7.1$d$), reflecting that decisions embed forecasts. Equations (6.7.1$c$) and (6.7.1$d$) assert that $d_t$ equals a fixed matrix $H$ times a vector of forecasts of discounted sums of current and future $y_t$ vectors, forecasts that equal appropriate conditional expectations (please see equation (2.4.17)).

[22] In applications, $A$ sometimes depends on deeper parameters.